

Mode-Matching Approach to Current Blocking Effect in Graphene Nanoribbons

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Zigzag graphene nanoribbon (ZGNR) p–n junctions display parity-dependent transport on the number of zigzag chains. We revisit this phenomenon using the mode-matching method and derive analytical solutions for the transmission probability. It is pointed out that the randomness of the interface tilting destroys the parity effect if the spread of the tilting parameter is larger than the lattice constant. We discuss the origin of the parity effect in connection with the lattice symmetry. Junctions with bearded ZGNRs are shown to strongly suppress electron transmission.

KEYWORDS: graphene nanoribbon, edge modes, mode-matching method

1. Introduction

Electrons in graphene behave as two-dimensional massless Dirac particles.^{1–3} They can tunnel completely through a region even of higher potential barrier than the incident energy (Klein paradox^{4–6}) and hence have a remarkably high mobility.⁷ This property is detrimental to achieving high on/off-ratio graphene-based digital transistors.⁸ However, zigzag graphene nanoribbons (ZGNRs), which are quasi-one-dimensional quantum wires of graphene, show that electron waves can be totally reflected by the potential barrier in the low-energy region, where the edge modes are responsible for the electronic conduction.⁹ This current-blocking effect (CBE), which stands in striking contrast to the Klein tunneling effect, may be utilized to confine electrons and also as a valley valve.¹⁰ It was found that the CBE sensitively depends on the parity of the number (N) of zigzag chains in ZGNRs.^{11,12}

In this study, we revisit the parity dependence of CBE in ZGNRs using a mode matching method. Figures 1(a) and 1(b) show the lattice structures of ZGNRs for even and odd N values, respectively. A finite potential barrier is applied for the semi-infinite region of $x_0 > 0$, as shown in Fig. 1(c). In the framework of the mode matching method, it is shown that the CBE can be understood as a result of the phase differences of the edge modes between the upper and lower edges. In addition, we derive the analytical expression of the transmission probabilities through the tilted interface for the generic wave vector k . The presence of randomness in interface tilting destroys the parity effect if the spread of the tilting parameter δ_0 becomes larger than the lattice constant. We also consider the origin of the parity effect in connection with the symmetry of the terminated honeycomb lattice. Scattering due to bearded edges is briefly discussed.

2. Energy Spectrum and Wavefunctions

Figure 2 shows the energy band structure of ZGNR as calculated using the nearest-neighbor tight-binding model,¹³ which is symmetric at $E = 0$ owing to particle–hole symmetry.⁶ A pair of central sub-bands, as highlighted by the thick black lines, is separated from the other sub-bands by a gap of Δ . The edge modes constitute the partially flat bands with $\pi/a \geq |k| > k_c$, where $k_c \approx 2\pi/3a$ for a large N . These modes are each characterized by a localization

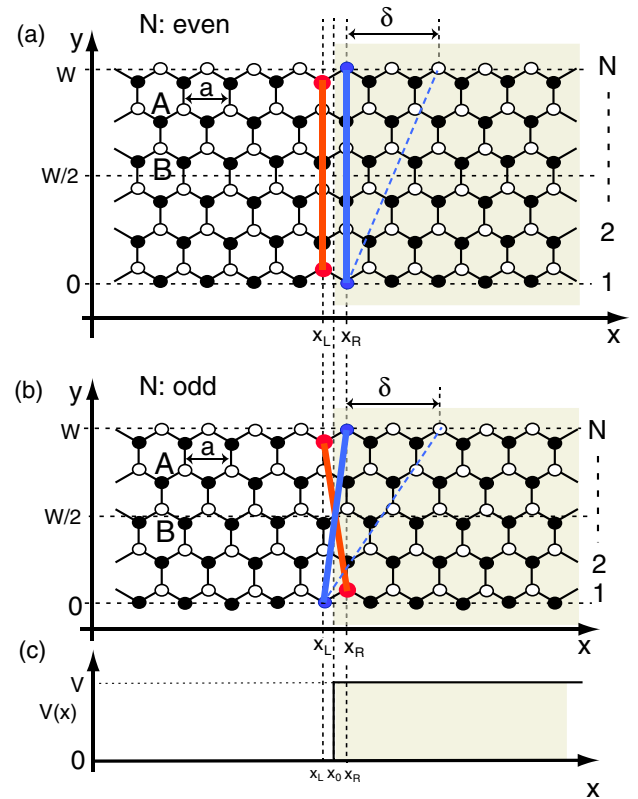


Fig. 1. (Color online) Schematics of ZGNRs consisting of (a) even and (b) odd numbers of zigzag chains, respectively. The black (white) circles represent A (B) sites. The $W = (a/2\sqrt{3})(3N - 2)$ denotes the ribbon width, with a being the lattice constant of graphene. In the shaded area, a gate is applied to create a vertical potential step as shown in (c). Wave functions have to be matched on lines $x = x_L$ and x_R . A mirror symmetry with respect to the mid-line $y = W/2$ exists in (a), while in (b) an inversion symmetry exists about the point $[x_0 = (x_L + x_R)/2, W/2]$. The potential breaks the inversion symmetry but preserves the mirror symmetry. In both (a) and (b), the dashed blue line indicates a tilted interface, as explained in the text.

length ξ_k , which tends to zero as k approaches π/a .¹⁴ The corresponding wave functions $\Psi_{sk} = [\psi_{mA}^s, \psi_{mB}^s]^T$ are labeled k and the particle–hole index $s = \pm$. Here, $m = 1, 2, \dots, N$ labels zigzag chains, as shown in Fig. 1. Neglecting a phase factor that is irrelevant for our purpose, they can be written as,¹⁴

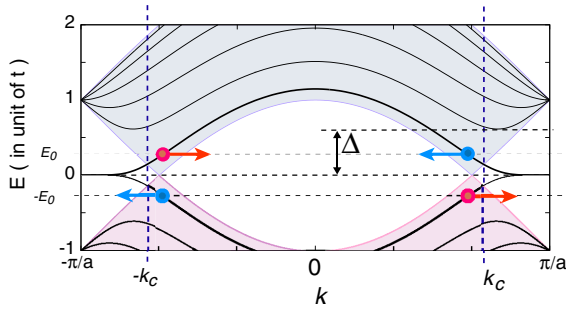


Fig. 2. (Color online) Band structure of ZGNRs with $N = 6$ with energy in unit of the nearest-neighbor hopping integral t . The edge modes have $|k| > k_c$ and $E \sim 0$. The arrows indicate the directions of group velocity. The band structures of ZGNRs with an odd number of chains look qualitatively the same.

$$\begin{aligned}\psi_{mA}^s(k, x) &= N_c e^{ikx} \sinh[\eta(N+1-m)], \\ \psi_{mB}^s(k, x) &= N_c e^{ikx} \sinh(\eta m),\end{aligned}\quad (1)$$

where $N_c = (1/\sqrt{2})[\sum_{m=1}^N \sinh^2(m\eta)]^{-1/2}$ is the normalization factor and the transverse quantum number η is determined by¹⁴⁾

$$\frac{\sinh(N\eta)}{\sinh[(N+1)\eta]} = g_k, \quad (2)$$

where $g_k = 2 \cos(ka/2)$. Note that η tends to infinity as k approaches π/a . The energy spectrum is given by¹⁴⁾

$$E = s\sqrt{1 + g_k^2 - 2g_k \cosh(\eta)}, \quad (3)$$

in units of the hopping integral t .

3. Mode-Matching

The scattering configuration is depicted in Fig. 1(c). A potential step $V > 0$ is created at $x = x_0 \equiv (1/2)(x_L + x_R)$ by superimposing a gate to the right region of $x \geq x_0$. The scattering is encoded by the wave function $\Psi(x, m)$ with the following asymptotic forms:

$$\begin{aligned}\Psi(x \rightarrow -\infty, m) &\rightarrow \Psi_{s=+,k} + R\Psi_{s=+,-k}, \\ \Psi(x \rightarrow \infty, m) &\rightarrow T\Psi_{s',k'},\end{aligned}\quad (4)$$

where Ψ_{sk} and $\Psi_{s'k'}$ represent right-going modes and $\Psi_{s,-k}$ represents a left-going mode, while R and T denote the reflection and transmission coefficients, respectively. We have assumed $s = +$, indicating a positive scattering energy $E > 0$. Note that $s' = \text{sgn}(E - V)$. In the vicinity of the interface $x = x_0$, however, there are also evanescent modes contributing to $\Psi(x, m)$ in addition to the propagating modes. In general,¹⁵⁾ there are $(N - 1)$ evanescent modes away from the interface on either side at a given energy $|E| < \Delta$. Therefore, we write $\Psi(x > 0, m) = \Psi_r(x, m)$ and $\Psi(x < 0, m) = \Psi_l(x, m)$, with $\Psi_r(x, m) = T\Psi_{s',k'} + \sum_v T_v \varphi_v^R(x, m)$ and $\Psi_l(x, m) = \Psi_{s,k} + R\Psi_{s,-k} + \sum_\mu R_\mu \varphi_\mu^L(x, m)$, where $\mu, v = 1, 2, \dots, N - 1$ tags the evanescent modes and $\varphi_v^R(x \rightarrow \infty, m) \rightarrow 0$ and $\varphi_\mu^L(x \rightarrow -\infty, m) \rightarrow 0$. It is seen that $2(N - 1)$ more coefficients (namely, T_v and R_μ) have been introduced in addition to R and T .

Therefore, there are $2N$ unknowns in total involved in the problem. In principle, they can be determined by matching $\Psi_r(x, m)$ to $\Psi_l(x, m)$ in such a manner that the Schrödinger

equation for the nearest-neighbor tight-binding model is satisfied throughout the sample. One can show that the necessary and sufficient conditions are that Ψ_r and Ψ_l match on the lines $x = x_R$ and x_L . Clearly, there are N sites on each line. Altogether, there are $2N$ sites; thus, these conditions yield $2N$ equations for the $2N$ coefficients. R and T can then be uniquely determined by solving these equations simultaneously.

3.1 Wide ribbon limit

Specifically, we consider $W/\xi_k \gg 1$. Under such circumstances, the wave function is negligible on all sites other than those on the peripheral lines $y = 0$ and W , as is evident from Eq. (1). From this observation, we infer that T_v and R_μ are also negligible, regardless of μ and v (see Appendix). Evanescent modes then become irrelevant.

R and T can be determined by satisfying the matching conditions on a pair of active sites, which are denoted as blue circles in Figs. 1(a) and 1(b). Let x_1 and x_N be the coordinates of the lower ($y = 0$) and upper ($y = W$) blue sites, respectively. For an even N , $(x_1, x_N) = (x_R, x_R)$ and for an odd N , $(x_1, x_N) = (x_L, x_R)$. Then, on the lower site, we have

$$\psi_{1A}^s(k; x_1) + R\psi_{1A}^s(-k; x_1) = T\psi_{1A}^{s'}(k'; x_1), \quad (5)$$

while on the upper site we have

$$\psi_{NB}^s(k; x_N) + R\psi_{NB}^s(-k; x_N) = T\psi_{NB}^{s'}(k'; x_N). \quad (6)$$

The substitution of Eq. (1) into Eqs. (5) and (6) yields

$$e^{ikx_N} + R e^{-ikx_N} = TC \cdot \frac{s'}{s} e^{ik'x_N}, \quad (7)$$

$$e^{ikx_1} + R e^{-ikx_1} = TC \cdot e^{ik'x_1}, \quad (8)$$

respectively. Here, $C = N'_c \sinh(N\eta')/N_c \sinh(N\eta)$ and (k, η) is determined at energy E through Eqs. (2) and (3), which also determine (k', η') , but with $E - V$ instead of E . We shall be mainly interested in cases where $|k| \sim |k'|$, namely, $V \sim 2E$ or $V \sim 0$, and hence $C \approx 1$.

3.2 Current blocking effect

We show that CBE occurs when N is even. In this case, $x_1 = x_N$. Subtracting Eq. (8) from Eq. (7), we immediately conclude that

$$\text{CBE: } T = 0, \text{ if } \frac{s'}{s} = -1, \quad (9)$$

which prohibits an electron from transmitting through a vertical potential step from a conduction (valence) band to a valence (conduction) band. Thus, we have obtained the CBE using the mode matching approach.

If $s = s'$, Eqs. (7) and (8) become identical and they have to be supplemented using an additional equation to uniquely determine R and T . This equation can be found by restoring the small amplitude on the red filled circles [see Fig. 1(a)] on the line $x = x_L$ and it reads $e^{ikx_L} + R e^{-ikx_L} = T e^{ik'x_L} C'$, where $C' = N'_c \sinh(\eta')/N_c \sinh(\eta)$. As with C , $C' \approx 1$. We then find $T \approx \sin(ka/2)/\sin[(k+k')a/4]$. Since $s'/s = 1$, we must have k and k' of the same sign. It follows that $T \approx 1$ for any $k \sim k'$.

However, CBE does not occur for an odd N . In this case, $x_1 - x_N = a/2$. From Eqs. (7) and (8), we find

$$T \approx \begin{cases} \frac{\sin\left(\frac{ka}{2}\right)}{\sin\left[\frac{(k+k')a}{4}\right]} & \text{if } \frac{s'}{s} = 1, \\ \frac{i \sin\left(\frac{ka}{2}\right)}{\cos\left[\frac{(k+k')a}{4}\right]} & \text{if } \frac{s'}{s} = -1. \end{cases} \quad (10)$$

In the case of $s'/s = 1$, we have k and k' of the same sign and then $T \approx 1$. On the other hand, in the case of $s'/s = -1$, k and k' must have opposite signs (see Fig. 2) and then $k + k' \sim 0$, leading to $T \approx i$ for k close to π/a . In either case, we have $|T|^2 \approx 1$, which indicates that electrons can perfectly transmit via edge modes even if the bands have been switched in contrast to an even ZGNR.

3.3 Tilted interface

As revealed above, the CBE can be attributed to the fact that the upper blue site has the same longitudinal phase as the lower one. By tilting the interface [see Figs. 1(a) and 1(b)], this situation is changed and the CBE may disappear. Suppose that the upper blue site is displaced by $\delta = na$ (n being an integer) relative to the lower one by tilting, then $x_N - x_1 = \delta$ for an even N . An additional phase $e^{ik\delta}$ then appears and we find from Eqs. (7) and (8) that¹⁶⁾

$$|T|^2 = \left| \frac{2 \sin(k\delta)}{\frac{s'}{s} e^{i\delta(k+k')/2} - e^{-i\delta(k+k')/2}} \right|^2. \quad (11)$$

This equation can be simplified if $s'/s = -1$ (and thus $k \sim -k'$) to $|T|^2 = (1/2)[1 - \cos(2k\delta)]$. In an analogous manner, one can show that $|T|^2 = (1/2)[1 + \cos(2k\delta)]$ for odd N and $k \sim \pm\pi/a$. Integrating them together, we find

$$|T|^2 = \frac{1}{2} [1 - (-1)^N \cos(2k\delta)], \quad (12)$$

which reproduces the results in Ref. 11 when $k = 2\pi/3a$.¹⁷⁾ The transmission probabilities for the tilted interface are shown in Fig. 3. In the presence of randomness in δ , which might occur in actual device fabrication, it may be interesting to look at the average transmission probability. Presuming a normal distribution for δ with a variation δ_0 and a zero mean, we find that

$$\langle |T|^2 \rangle = \frac{1}{2} \left[1 - (-1)^N \exp\left(-\frac{k^2 \delta_0^2}{2}\right) \right], \quad (13)$$

where $\langle \cdots \rangle$ indicates the average. Figure 3(c) shows the averaged transmission probabilities for ZGNRs with even and odd N values. It can be seen that the average transmission probabilities converge to $1/2$ in both cases with an increase in $\delta_0 \sim a$, i.e., with the disappearance of the parity effect.

4. Symmetry Analysis

We discuss the symmetries of even and odd ZGNR lattices to explain the CBE. As shown in Fig. 1(a), a ZGNR with an even number of zigzag chains has a mirror symmetry at $y = W/2$. The nearest-neighbor tight-binding Hamiltonian

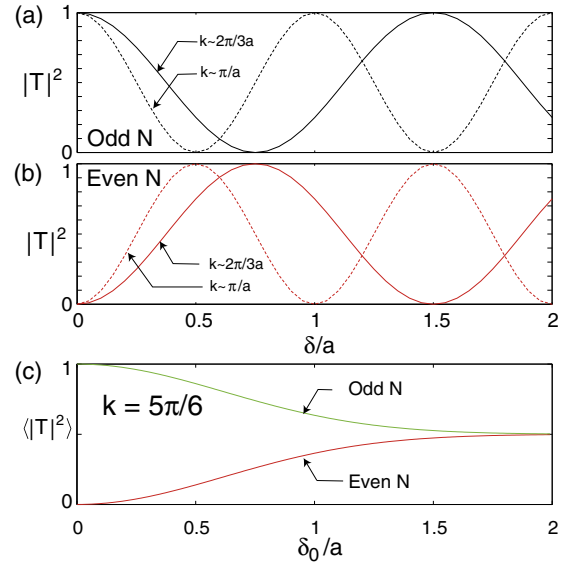


Fig. 3. (Color online) Transmission probability $|T|^2 = (1/2)[1 - (-1)^N \cos(2k\delta)]$ vs the interface tilting parameter δ at different k values for (a) even and (b) odd N values. (c) Average transmission probabilities $\langle |T|^2 \rangle = (1/2)[1 - (-1)^N \exp(-k^2 \delta_0^2/2)]$ vs the variation δ_0 .

is invariant under the operation $\hat{R} : y_i \rightarrow W - y_i$, which entails a parity denoted by $P = 0, 1$ in the Hamiltonian eigenstates. The step potential [$V(x > x_0) = V$ and $V(x < x_0) = 0$] is also invariant under $\hat{R}$, from which it immediately follows that $V(x)$ cannot change the parity during scattering. Acting $\hat{R}$ on the edge mode simply changes mA to $(N + 1 - m)B$ and simultaneously mB to $(N + 1 - m)A$. One then infers [from Eq. (1)] that the parity for the edge mode is $(-1)^{P+1} = s$. Thus, $\langle \Psi_{s,k} | V(x) | \Psi_{s',k'} \rangle = 0$ if $s \neq s'$, as concluded from the results of the wave function matching method.

On other hand, a ZGNR with an odd number of zigzag chains does not have a mirror symmetry. Instead, it features an inversion center at $(x_0, W/2)$. The Hamiltonian is invariant under $\hat{I} : (x_0 + x, y) \rightarrow (x_0 - x, W - y)$. Similar to the case of a ZGNR with an even number of zigzag chains, one can also assign a definite parity ($I = 0, 1$) to the eigenstates, which for the edge mode is given by $(-1)^{I+1} = s$. However, such a parity can be changed in the course of scattering by $V(x)$, since $V(x)$ is not invariant under $\hat{I}$. The inversion symmetry of an odd ZGNR is broken by the presence of a step potential.

The above symmetry is also evident in the matching conditions. Owing to the $\hat{R}$ symmetry, the longitudinal phases on the two active sites in a ZGNR with an even number of zigzag chains are the same; however, owing to the $\hat{I}$ symmetry, they differ in a ZGNR with an odd number of zigzag chains. The $\hat{R}$ symmetry of $V(x)$ indicates that, the wave functions $(\Psi_{r,l})$ on both sides of the interface must have the same parity P ; otherwise they can not fulfill the matching conditions. The $\hat{R}$ symmetry is preserved by any potential profile that is symmetric about $y = W/2$. Therefore, we expect the CBE to occur within the single-mode regime in ZGNRs with an even number of zigzag chains in the presence of any $\hat{R}$ -invariant $V(x, y)$ as long as $E \cdot [E - V(x \rightarrow \infty, y)] < 0$ and $V(x, y)$ approaches a constant value as $x \rightarrow \infty$.

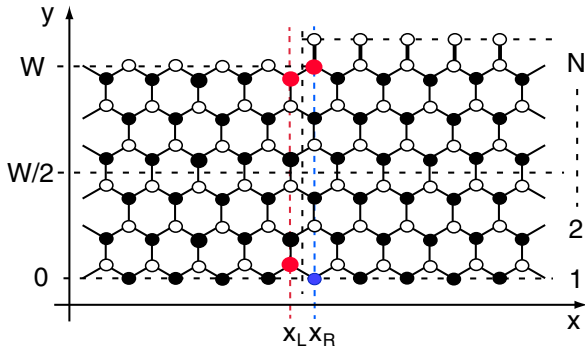


Fig. 4. (Color online) Sketch of junctions with bearded ZGNR. The circles on $y = W + a/\sqrt{3}$ represent bearded sites, where wave functions have the biggest amplitude. The transmission is strongly suppressed, $|T|^2 \approx 0$, in the edge mode regime.

4.1 Zigzag-bearded junctions

Finally, we comment on the scattering due to the presence of bearded edges (without applying a voltage, see Fig. 4). Since a bearded ZGNR ($x_R > 0$) becomes an insulator, the junction is a metal–insulator junction,¹⁸⁾ which gives a perfect reflection. Here, we show that such a reflection can also be understood as a result of the matching conditions. The transmission coefficient is obtained by matching the wave functions on the sites situated at $(x_R, 0)$ and (x_R, W) , respectively. On these sites, the corresponding Ψ_l is $\propto e^{ikx_R} + R e^{-ikx_R}$. On other hand, Ψ_r is around zero at (x_R, W) because of the nodal site,¹⁹⁾ but $\Psi_r \propto T e^{ikx_R}$ at $(x_R, 0)$. By equating them individually, we obtain $|T|^2 \approx 0$, indicating that transmission is prohibited. Note that the prohibition is not due to any selection rules, since neither $\hat{R}$ nor $\hat{I}$ symmetry exists in the junction. Rather, it originates from the fact that bearded sites cause a strong localization and the depletion of amplitude at $y = W$ sites.

5. Conclusions

In conclusion, we have shown the CBE in ZGNRs using the mode matching method. We have also derived the analytical expression of the transmission probability $|T|^2$ for the tilted interface for generic longitudinal wavevectors on the basis of this method. The randomness of interface tilting destroys the parity effect if the spread of the tilting parameter δ_0 becomes larger than the lattice constant. The CBE can be attributed to the lattice symmetry of ZGNRs. Junctions with bearded ZGNRs have been shown to strongly suppress the electron transmission irrespective of the lattice symmetry.

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Appendix

In this appendix, we prove that T_v and R_μ are

insignificant. Since the odd- N case can be proved in a similar manner to the even- N case, in what follows we shall take the latter as an example.

For convenience, let us label the $2N$ boundary sites (as indicated in Fig. 1) by a single number $i = 1, 2, \dots, 2N$. The corresponding coordinates are then given as (x_i, m_i) , where $x_i = x_{L,R}$ and $m_i = 1, \dots, N$. For clarity, we shall arrange it in such a way that $i = 1$ and 2 correspond to the lower and upper blue sites, respectively, i.e., $(x_1, m_1) = (x_R, 1)$ and $(x_2, m_2) = (x_R, N)$, while $i = 3, \dots, N+2$ correspond to $(x_L, m_i = i-2)$ and $i = N+3, \dots, 2N$ to $(x_R, m_i = i - (N+1))$. Moreover, $\Psi_{s,k}(i) \equiv \psi_{m_i,A}^s(k, x_i)$ if this site belongs to the A-sublattice, while $\Psi_{s,k}(i) \equiv \psi_{m_i,B}(k, x_i)$ otherwise. Now using the matching conditions, we obtain a set of $2N$ equations as follows:

$$\Psi_r(i) = \Psi_l(i), \quad i = 1, \dots, 2N,$$

where $\Psi_{r/l}(i) \equiv \Psi_{r/l}(x_i, m_i)$. After inserting the expressions of $\Psi_{l/r}$ (see Sect. 3), the above equations can be explicitly written as

$$\begin{aligned} \Psi_{s,k}(i) + R \Psi_{s,-k}(i) + \sum_{\mu} R_{\mu} \varphi_{\mu}^L(i) \\ = T \Psi_{s',k'}(i) + \sum_v T_v \varphi_v^R(i). \end{aligned}$$

Here, note that φ_{μ}^L and φ_v^R are the left- and right-going evanescent modes, respectively. Rearranging the terms, we arrive at

$$\begin{aligned} T \Psi_{s',k'}(i) - R \Psi_{s,-k}(i) + \sum_v T_v \varphi_v^R(i) - \sum_{\mu} R_{\mu} \varphi_{\mu}^L(i) \\ = \Psi_{s,k}(i). \end{aligned}$$

Note that these equations can be transformed in matrix form, namely,

$$\sum_j M(i,j) V_j = C_i,$$

where $M(i,j)$ denotes the value of the j -th mode at the i -th boundary site, $V_j = T, R, T_v, R_{\mu}$ and $C_i = \Psi_{s,k}(i)$. For an even N , we have $C_i = N_c e^{ikx_R} f_i \sinh(\eta y_i)$, where $y_i = N + 1 - m_i$ and $f_i = s$ for A sites, whereas $y_i = m_i$ and $f_i = 1$ for B sites. To distinguish $[T, R]$ from $[T_v, R_{\mu}]$ as well as the blue sites from interior boundary sites, we break down the matrices in blocks:

$$M = \begin{pmatrix} S & Q \\ P & U \end{pmatrix}, \quad V = \begin{pmatrix} V_s \\ V_u \end{pmatrix}, \quad C = \begin{pmatrix} C_s \\ C_u \end{pmatrix}$$

where $C_s = [C_1, C_2]^T$, $V_s = [T, R]^T$, $C_u = [C_3, \dots, C_{2N}]^T$, and $V_u = [T_v, R_{\mu}]^T$, so that we get

$$\begin{aligned} S V_s + Q V_u &= C_s, \\ P V_s + U V_u &= C_u. \end{aligned}$$

Clearly, S , Q , P , and U are matrices of 2×2 , $2 \times 2(N-1)$, $2(N-1) \times 2$, and $2(N-1) \times 2(N-1)$ dimensions, respectively. For completeness, we also write them down explicitly. S is expressed as

$$S = \begin{pmatrix} \psi_{1A}^{s'}(k', x_R) & -\psi_{1A}^s(-k, x_R) \\ \psi_{NB}(k', x_R) & -\psi_{NB}(-k, x_R) \end{pmatrix},$$

while U is expressed as

$$U = \begin{pmatrix} \varphi_1^R(3) & \dots & \varphi_{N-1}^R(3) & -\varphi_1^L(3) & \dots & -\varphi_{N-1}^L(3) \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ \varphi_1^R(2N) & \dots & \varphi_{N-1}^R(2N) & -\varphi_1^L(2N) & \dots & -\varphi_{N-1}^L(2N) \end{pmatrix}.$$

Additionally, we have

$$P = \begin{pmatrix} \Psi_{s',k'}(3) & -\Psi_{s,-k}(3) \\ \vdots & \vdots \\ \Psi_{s',k'}(2N) & -\Psi_{s,-k}(2N) \end{pmatrix},$$

and

$$Q = \begin{pmatrix} \varphi_1^R(1) & \dots & \varphi_{N-1}^R(1) & -\varphi_1^L(1) & \dots & -\varphi_{N-1}^L(1) \\ \varphi_1^R(2) & \dots & \varphi_{N-1}^R(2) & -\varphi_1^L(2) & \dots & -\varphi_{N-1}^L(2) \end{pmatrix}.$$

In the wide ribbon limit, from Eq. (2) one deduces that $e^{-\eta} \approx g_k$ for $(2\pi/3a + 1/W) < k \leq \pi/a$. In particular, η tends to infinity as k approaches π/a . Thus, $|C_i/C_1| = |C_i/C_2| \approx e^{-\eta\delta y_i}$, where $\delta y_i = m_i - 1$ for A sites and $\delta y_i = N - m_i$ for B sites, and it measures the distance of A (B) sites from the lower (upper) edge. From this, we see that the elements of P and C_u are much smaller than those of S and C_s . Such smallness is characterized by the parameter $\theta = e^{-\eta} \ll 1$. Now that both C_u and PV_s are on the order of θ , we then deduce that $V_u = U^{-1}(C_u - PV_s)$ must also be on the order of θ , i.e., T_v and R_μ are insignificant. Hence, QV_u , which is then also on the order of θ , is negligible in comparison with C_s . Therefore, V_s is determined approximately by

$$SV_s \approx C_s$$

which yields Eqs. (5) and (6). In other words, S essentially approximates the scattering matrix of the system.

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16) Note that, in the case of a tilted interface, one generally has to match the same set of modes on the same number of sites as in the case of the vertical one, but the sites are not the same any more.

17) Actually, with $\delta = na$, we have $|T|^2 = (1/2)[1 - (-1)^N \cos(4\pi n/3)] = (1/2)[1 - (-1)^N \cos(2\pi n/3)]$, which gives Eq. (2) in Ref. 11 when δ is plugged back.

18) K. Wakabayashi: *Phys. Rev. B* **64** (2001) 125428.

19) K. Wakabayashi, S. Okada, R. Tomita, S. Fujimoto, and Y. Natsume: *J. Phys. Soc. Jpn.* **79** (2010) 034706.